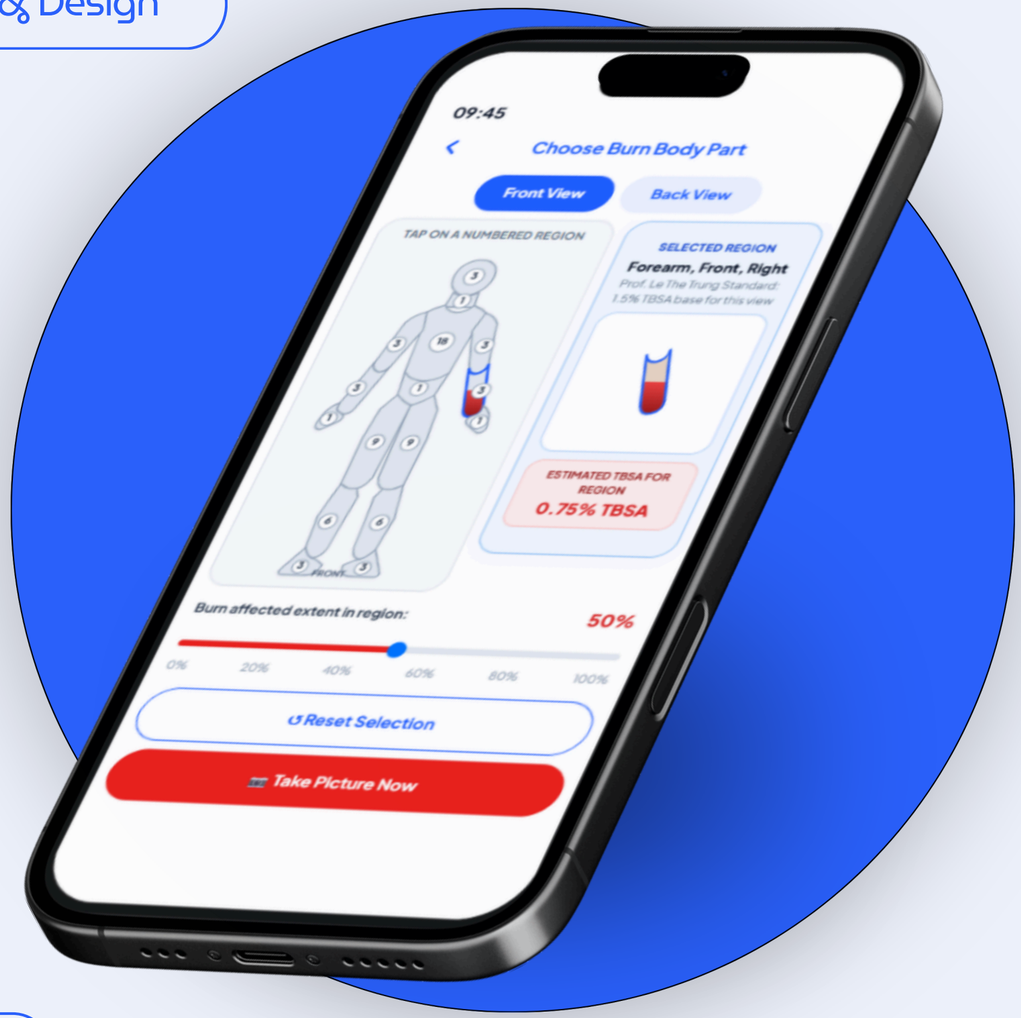
UEH UNIVERSITY

INSTITUTE OF INTELLIGENT & INTERACTIVE TECHNOLOGIES

Information System Analysis & Design

# BURN AID

*Burn assessment & First-aid guidance Information system*





GROUP 3

Đức Huy - Phương Thảo - Quốc Túy - Khánh Uyên - Lưu Vũ

Burn Aid

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# Background & Problem

- Burn injuries are a **public health concern** that can occur at home, in the workplace, and in other everyday settings.
- Before medical help arrives, patients must j**udge severity and give first aid**.
- AI and computer vision can **classify burn depth and outline wound**s from photos.

How might we **provide preliminary burn assessment** to whom experience burn before medical help is available **to support decision-making**.

# Proposed Solution

## *Burn Assessment and First Aid Guidance Information System*

1. Upload burn photo + short survey
2. AI classifies depth, segments burn
3. First-aid guidance

### Objective

Develop and evaluate an integrated prototype that supports early burn assessment and access to reliable first-aid information

# Scope

## In scope

- Adults aged 18+
- Image + survey assessment
- First-aid guidance and warnings

## Out of scope

- Diagnosis or treatment
- Children under 18
- Ambulance or location sharing

# Stakeholders

| Stakeholders | Needs | Influence Level |
|---|---|---|
| Department of Health, public health organizations | • Reduce the burden on hospital emergency departments from minor burn cases<br>• Promote standardized public first-aid practices. | High |
| Patient and Family members | • Receive appropriate assessment and first-aid guidance for minor burns and timely emergency assistance for severe burns. | High |
| Medical advisor | • Ensure that first-aid guidance follows medical standards and validate the reliability of the AI model. | High |
| IT Staff | • Ensure stable system performance and protect users' personal image data. | Medium |
| Emergency services | • Receive emergency phone calls initiated by users via the device's native dialer when recommended by the app. | Low |

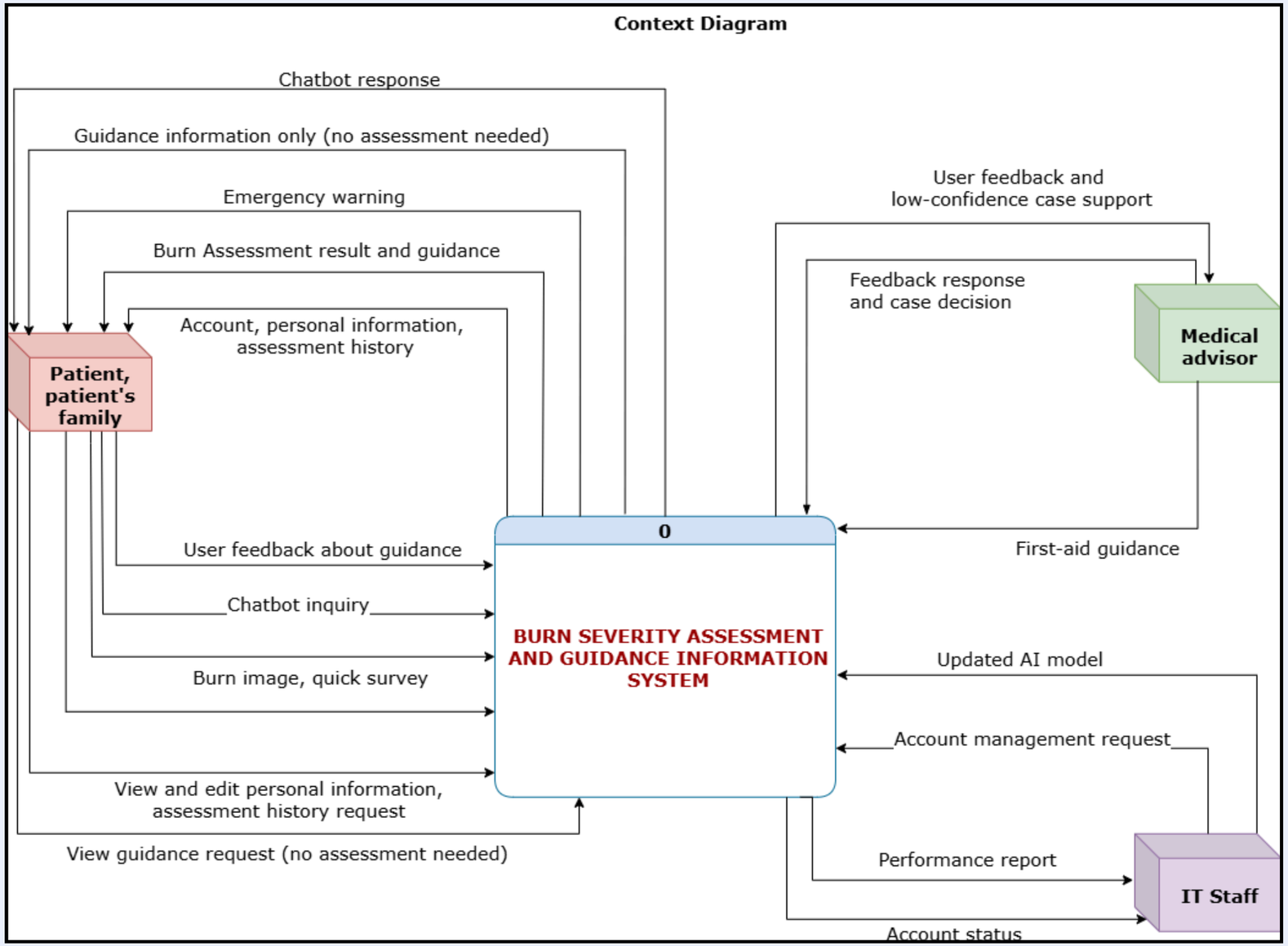
# User roles and requirements

| User roles | Description |
| --- | --- |
| Patient and Family member | • Primary user group of the system.<br>• Use it when they need a quick preliminary assessment of a burn, suspect a burn injury, or require appropriate first-aid guidance. |
| Medical Advisor | • Reviewing and managing professional first-aid content within the system. |
| IT Staff | • Managing the AI-based assessment models and maintaining the system.<br>• Ensure that the system functions, user accounts, and model-generated assessments operate appropriately and effectively. |

# Context Diagram

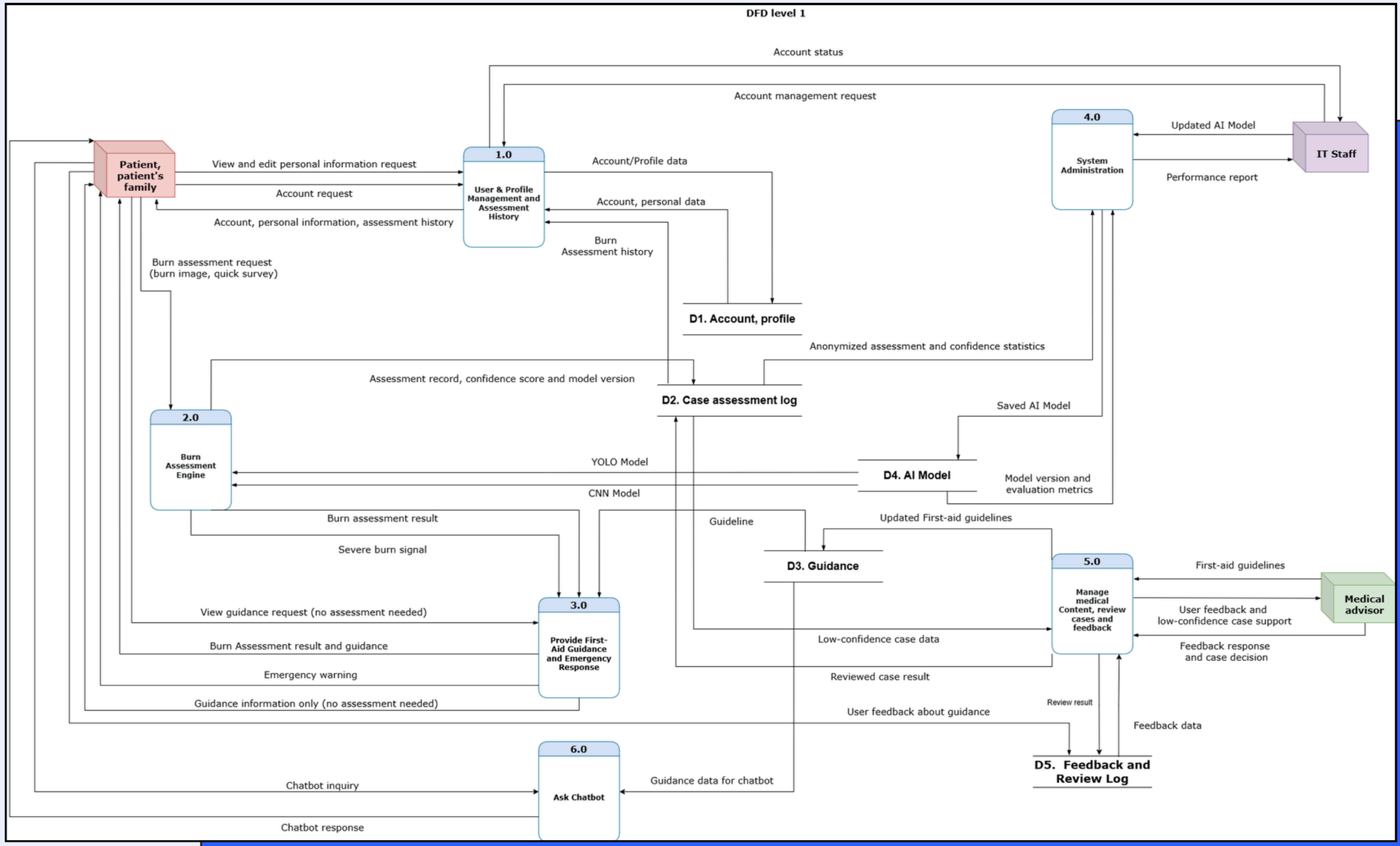
3 external entities:

- Patient and Family member;
- Medical Advisor;
- IT Staff




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# DFD *level 1*



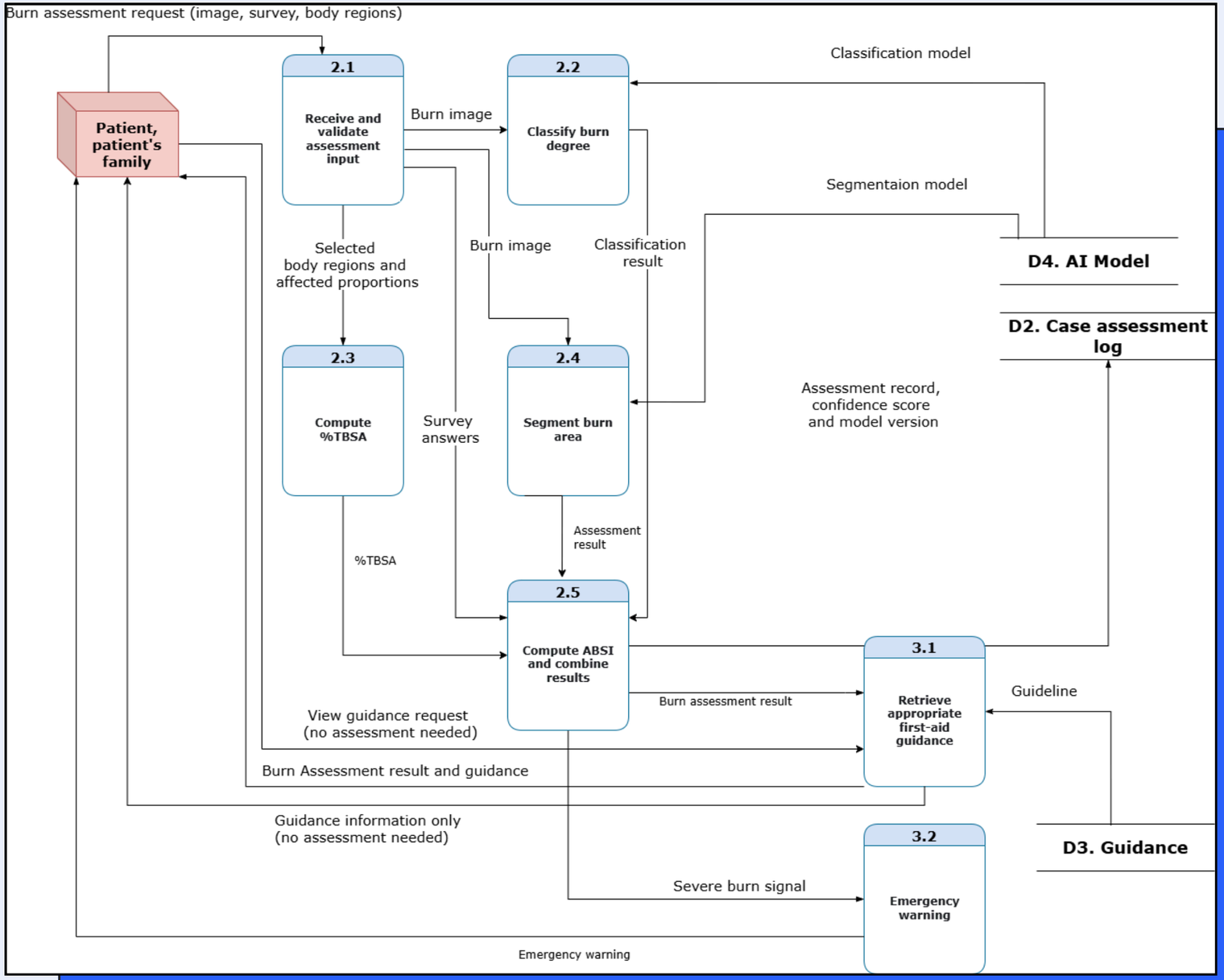

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# DFD level 2

**Process 2.0** has
5 sub-processes.

**Processes 3.0:**

- Guidance, emergency warning,
- Case review.

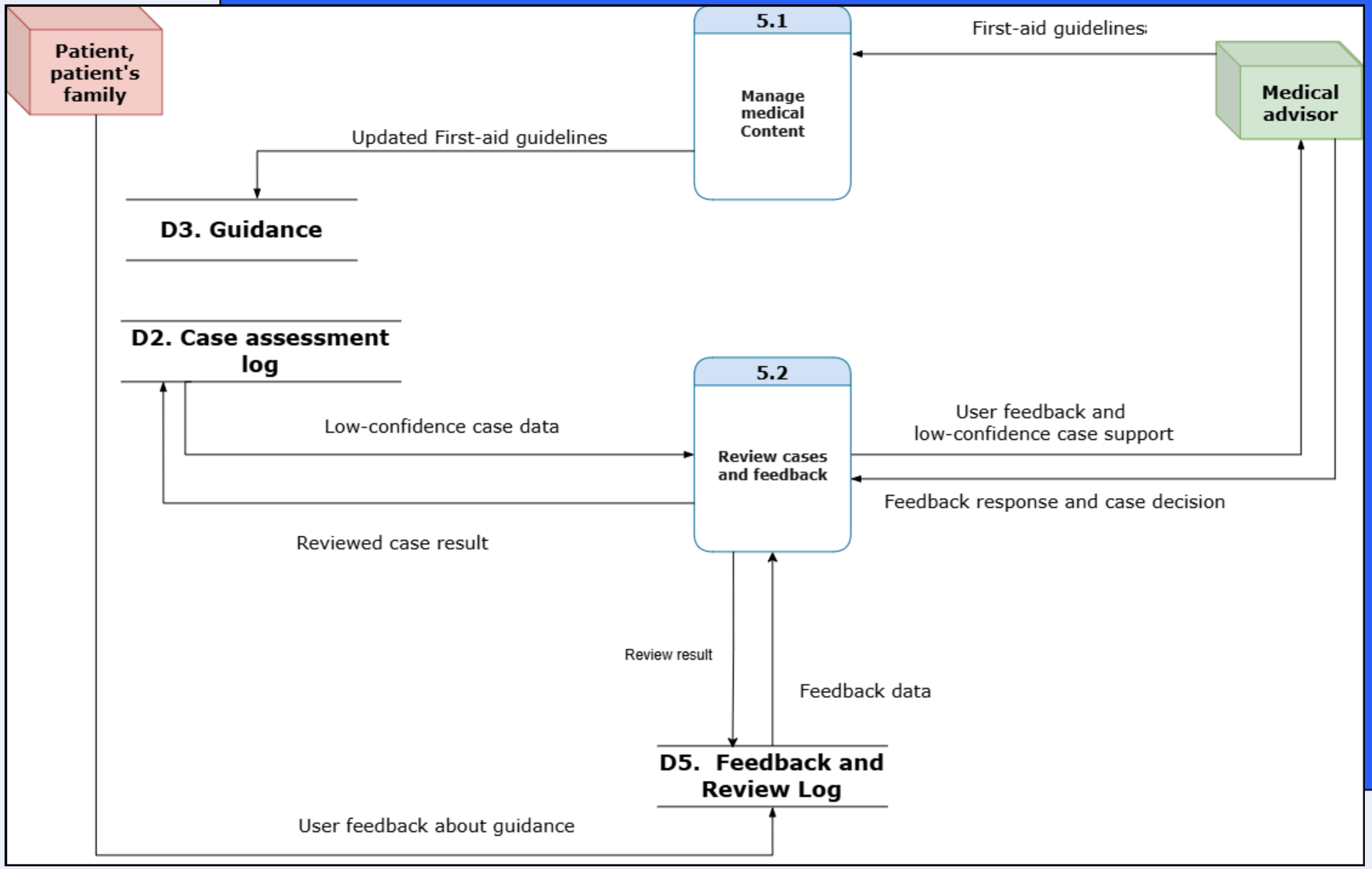



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# DFD level 2

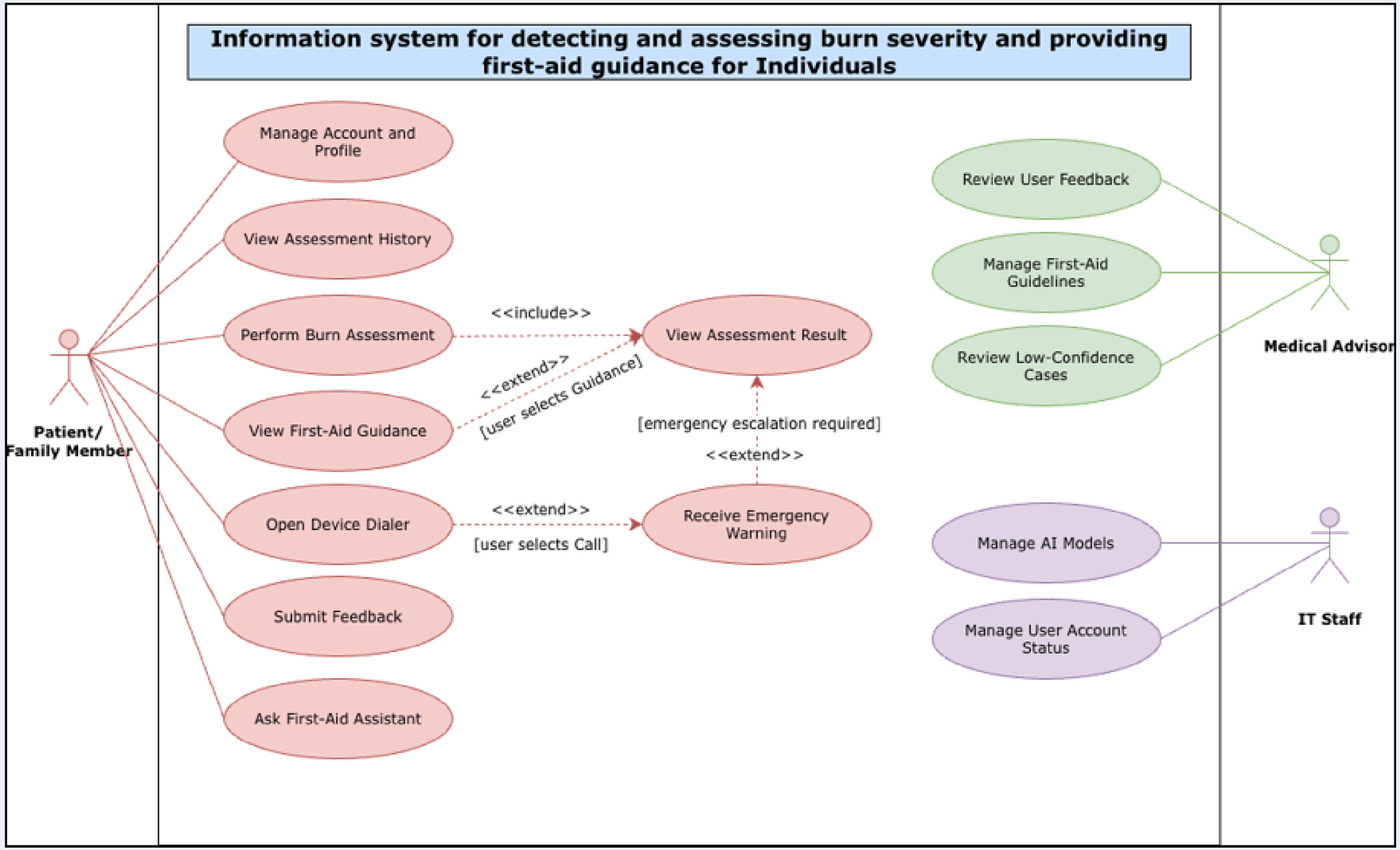
**Processes 5.0:**

- Manage medical review,
- Review cases and feedback.




Burn Aid

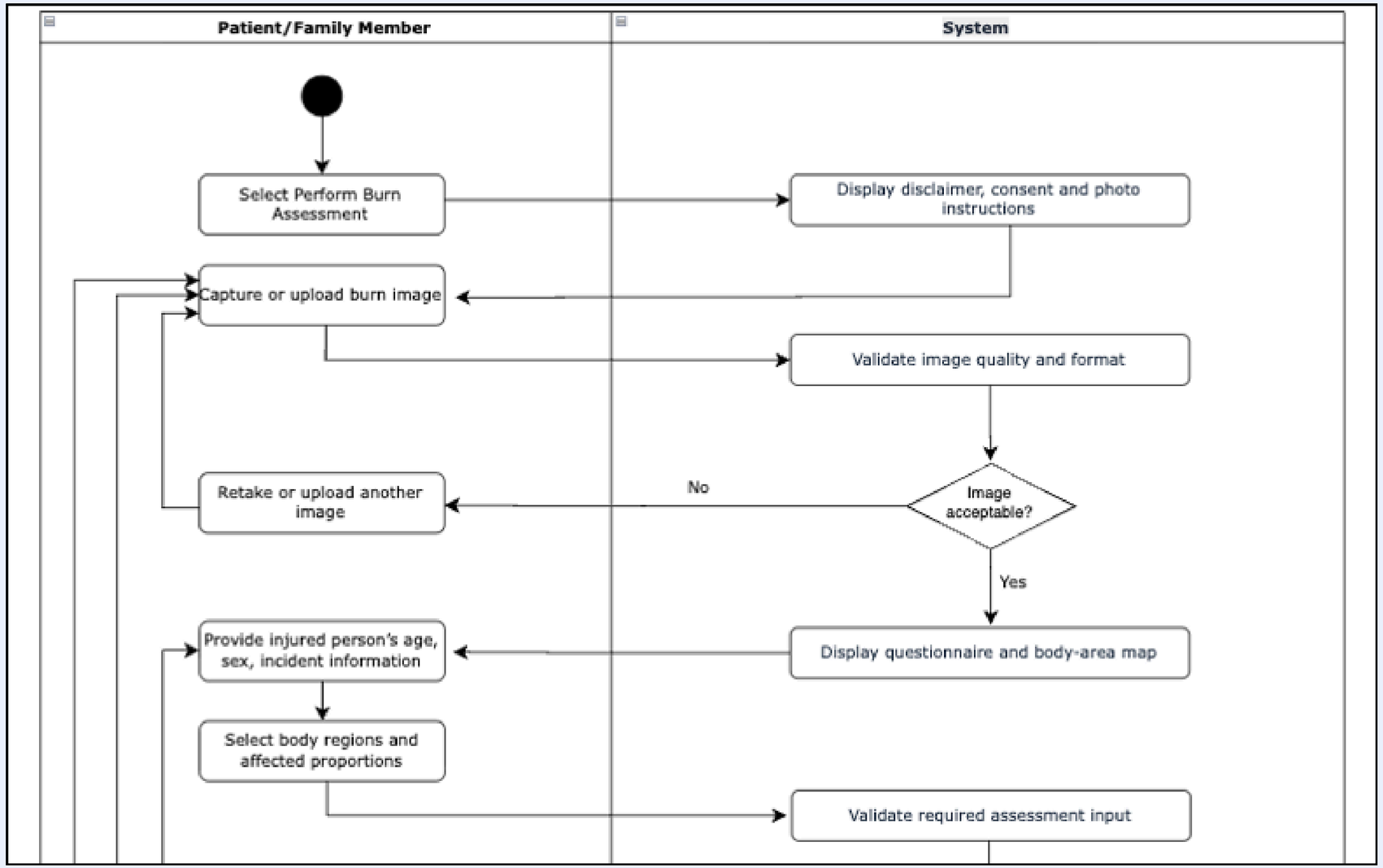
# Use Case Diagram




| ID | Use Case |
|---|---|
| UC-01 | Manage Account and Profile |
| UC-02 | View Assessment History |
| **UC-03** | **Perform Burn Assessment** |
| UC-04 | View Assessment Result |
| UC-05 | View First-Aid Guidance |
| UC-06 | Receive Emergency Warning |
| UC-07 | Open Device Dialer |
| UC-08 | Submit Feedback |
| UC-09 | Review User Feedback |
| UC-10 | Manage First-Aid Guidelines |
| UC-11 | Review Low-Confidence Cases |
| UC-12 | Manage AI Models |
| UC-13 | Manage User Account Status |
| UC-14 | Ask First-Aid Assistant |

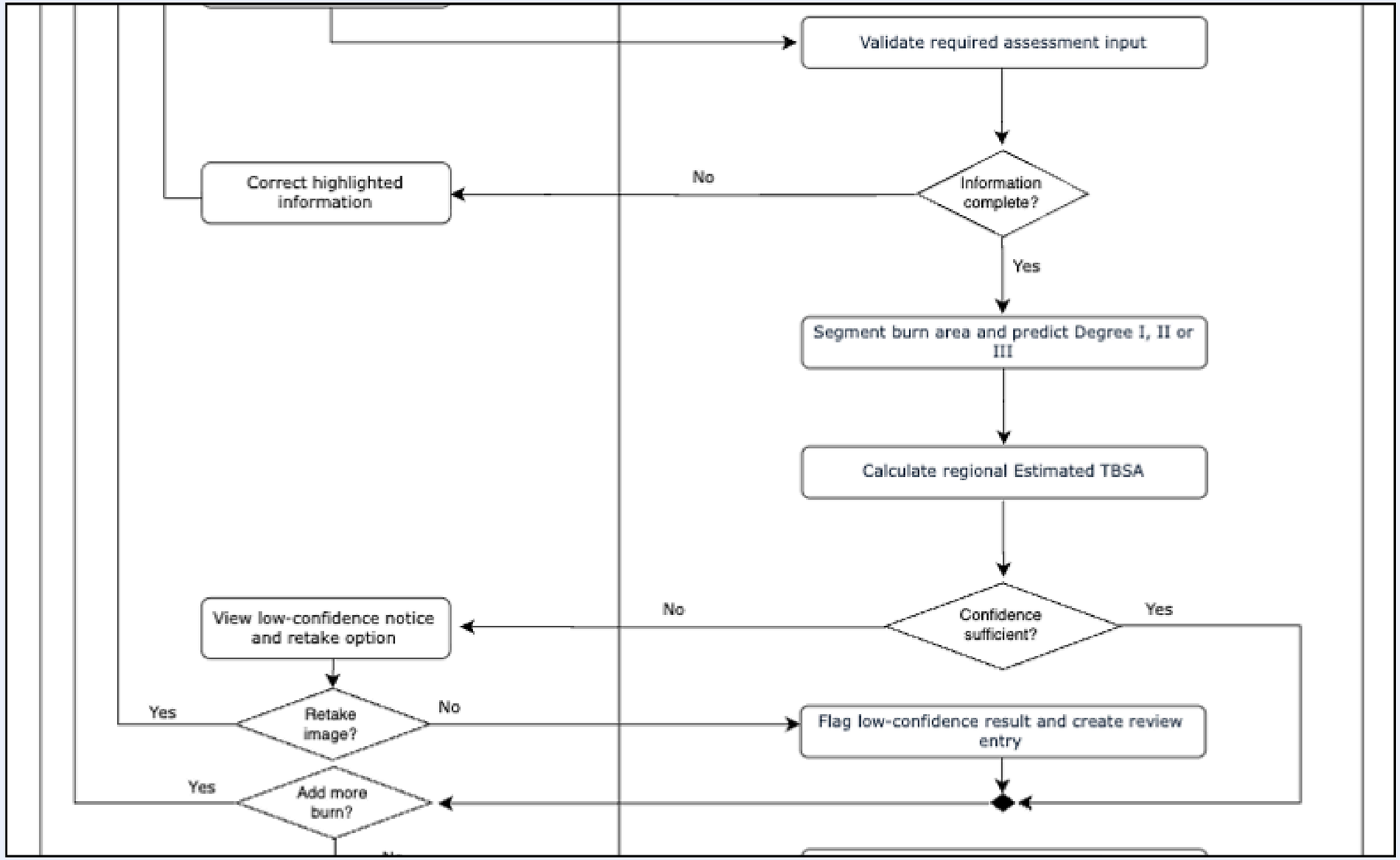
# Activity Diagram for UC-03

*Perform Burn Assessment*



# Activity Diagram for UC-03

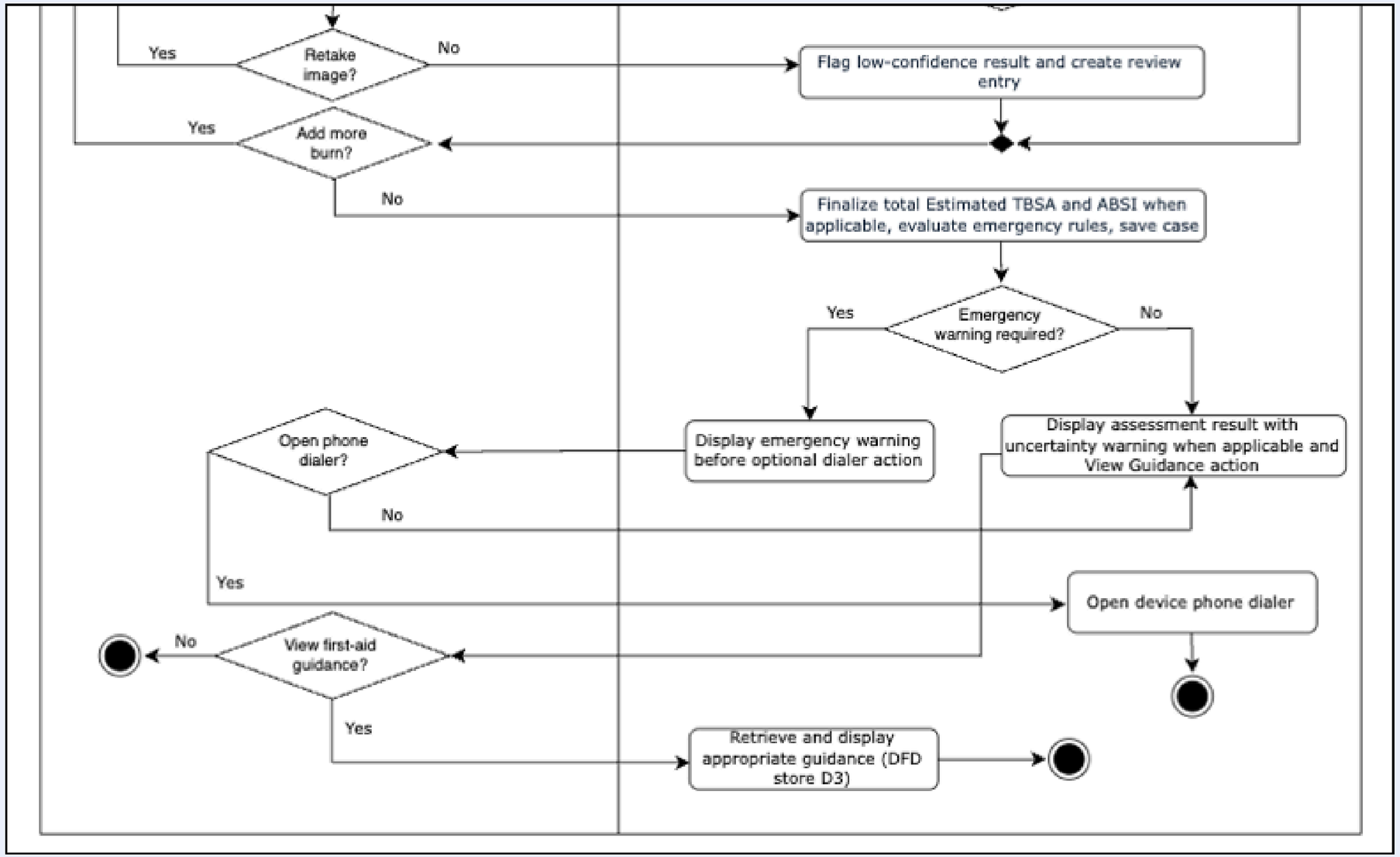
*Perform Burn Assessment*




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# Activity Diagram for UC-03

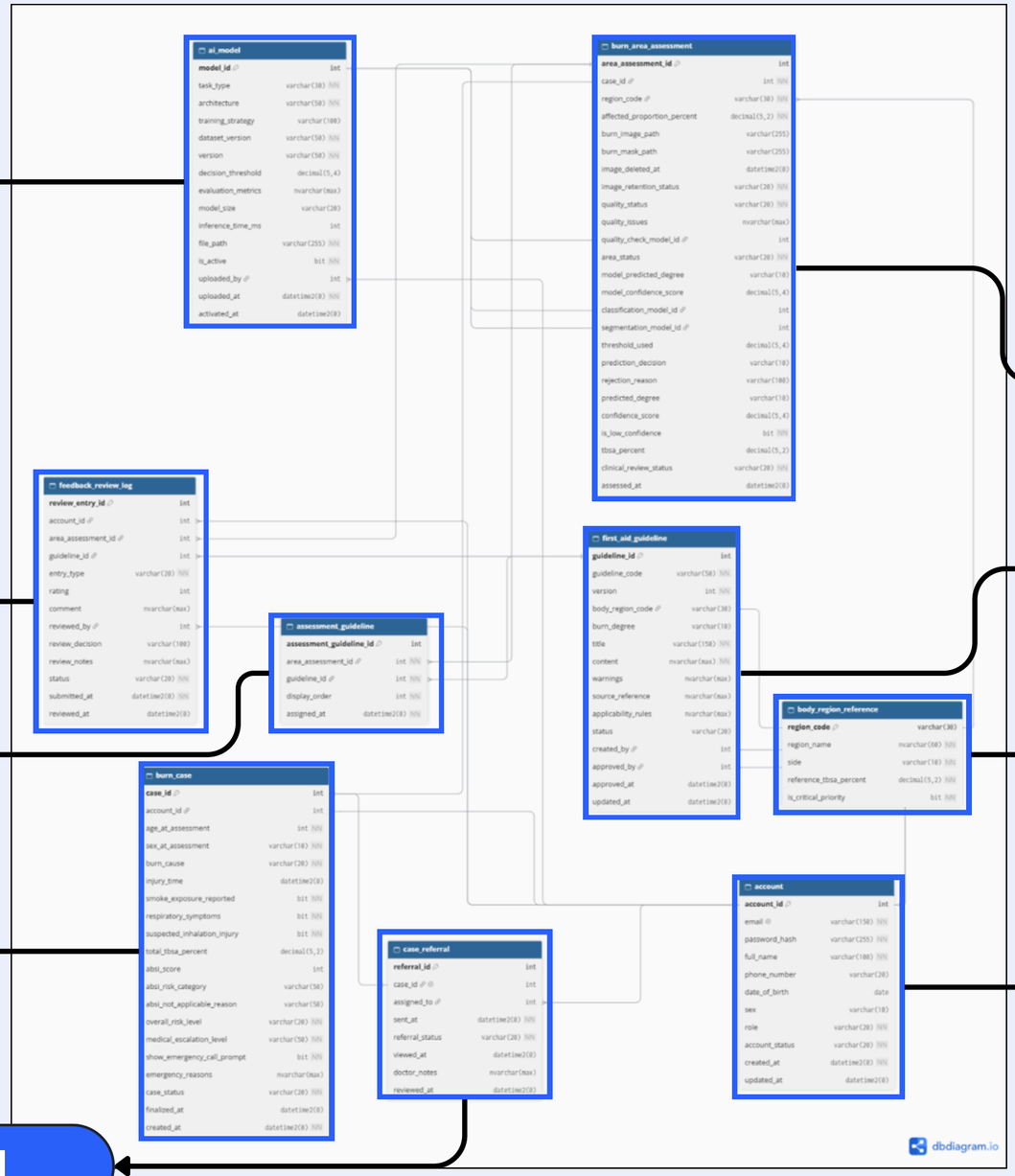
*Perform Burn Assessment*



# Relational Schema

| Table | Relational Schema |
|---|---|
| *Account* | Account(**account_id**, email, full_name, role, account_status, ...) |
| *Burn_case* | Burn_case(**case_id**, account_id (FK), age_at_assessment, burn_cause, case_status, ...) |
| *Burn_area_assessment* | Burn_area_assessment(**area_assessment_i**d, case_id (FK), region_code (FK), predicted_degree, confidence_score, tbsa_percent, ...) |
| *Body_region_reference* | Body_region_reference(**region_code**, region_name, side, standard_tbsa_percent, ...) |
| *First_aid_guideline* | First_aid_guideline(**guideline_id**, created_by (FK), guideline_code, version, title, status, ...) |
| *Assessment_guideline* | Assessment_guideline(**assessment_id**, area_assessment_id (FK), guideline_id (FK), display_order, ...) |
| *AI_model* | AI_model(**model_id**, task_type, architecture, version, is_active, ...) |
| *Feedback_review_log* | Feedback_review_log(**log_id**, case_id (FK), area_assessment_id (FK), guideline_id (FK), account_id (FK), entry_type, status, ...) |
| *Case_referral* | Case_referral(**referral_id**, case_id (FK), assigned_to (FK), referral_status, viewed_at, ...) |

# ERD




AI_model

Feedback_review_log

Assessment_guideline

Burn_case

Case_referral

Burn_area_assessment

First_aid_guideline

Body_region_reference

Account

Burn Aid

# Model Development

***Objective***

Support pre-hospital first aid for thermal burns by classifying burn depth and locating suspected burn regions.

| | | |
|---|---|---|
| **Burn depth clasification** | **ConvNeXtV2 Nano · 224 × 224 RGB** | Image-level probabilities for four classes: First-degree, Second-degree, Third-degree, and None (no burn) |
| **Burn region segementation** | **U-Net++ with ResNet-34 · 320 × 320 RGB** | Binary burn mask and visual overlay identifying suspected burn regions |
| **Clinical assessment** | **Independent Clinical Rule Engine** | Estimated %TBSA, risk level, and emergency alerts |

***Inference flow:*** *Image → Classification → If burn detected: Segmentation → Clinical assessment*

# Classification model

## Dataset

4,686 RGB images across four relatively balanced classes. ***Observed split:*** 83.40% train / 9.86% validation / 6.74% test

| Class | Train | Validation | Test | Total |
|---|---|---|---|---|
| First-degree | 1,076 | 110 | 73 | 1,259 |
| Second-degree | 1,021 | 97 | 83 | 1,201 |
| Third-degree | 834 | 140 | 81 | 1,055 |
| No Burn (None) | 977 | 115 | 79 | 1,171 |
| **Total** | 3,908 | 462 | 316 | 4,686 |

# Classification model

## Data cleaning and preprocessing

### Cleaning & labels

- Crop and visually filter images in Roboflow
- Manually standardize the four class labels
- Screen duplicates using pHash similarity
- Check image files and single-label validity

**All splits: RGB → tensor → ImageNet normalization**

Standardize image quality and labels before fine-tuning

### Image transforms

- Train: RandomResizedCrop to 224 × 224
- Horizontal / vertical flips; rotation ±15°
- ColorJitter for lighting variation
- Validation / test: fixed resize to 224 × 224

# Classification model

## Training setup

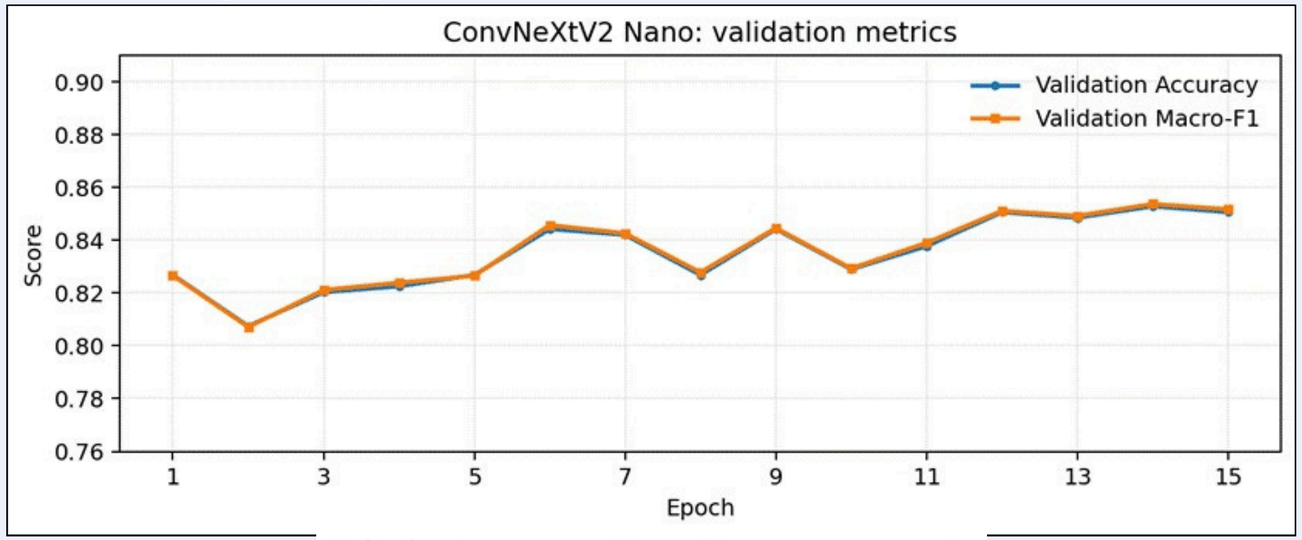
the ConvNeXtV2 Nano architecture via the timm library, initialized with pretrained weights convnextv2_nano.fcmae_ft_in22k_in1k

| Setting | Configuration |
|---|---|
| Architecture / initialization | ConvNeXtV2 Nano · 14.99M parameters · pretrained |
| Training budget | 15 epochs · Batch size 64 |
| Optimizer | AdamW · Weight decay 0.05 |
| Learning rates | Backbone: $2 \times 10^{-4}$ \| Head: $2 \times 10^{-3}$ |
| Scheduler | OneCycleLR · pct_start = 0.15 |
| Loss / regularization | Cross-entropy · Label smoothing 0.05 · Dropout 0.4 |

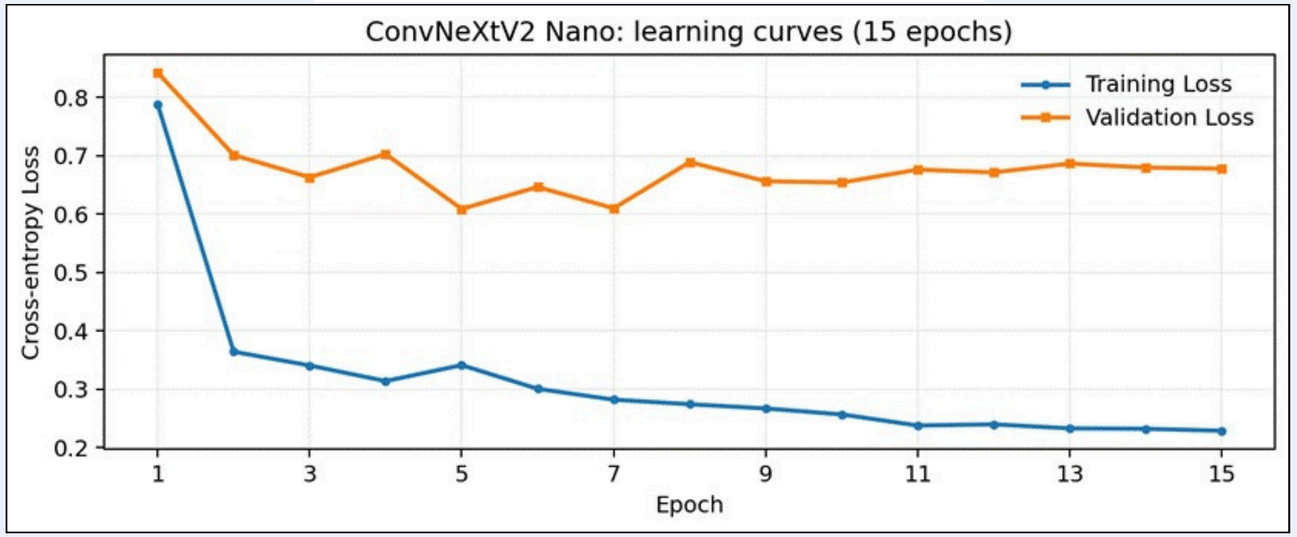
**Checkpoint selection: highest validation Macro-F1**

## Training Progress

Lower training loss does not guarantee better validation performance




*Validation Accuracy and Macro F1-score across epochs*




## Classification model

# EPOCH 14

Selected checkpoint

**85.37%** Macro-F1

**85.28%** Accuracy

Training loss

**0.7878 → 0.2285**

**Validation loss fluctuates; monitor overfitting.**

# Classification model

## Results on test split

Best checkpoint evaluated on 316 held-out images

**85.76%**
Test accuracy

**85.51%**
Test Macro-F1

| Class | Precision | Recall | F1-score |
|---|---|---|---|
| First-degree | 85.00% | 69.86% | 76.69% |
| Second-degree | 88.31% | 81.93% | 85.00% |
| Third-degree | 73.00% | 90.12% | 80.66% |
| No Burn (None) | 100.00% | 100.00% | 100.00% |
| Macro average | 86.58% | 85.48% | 85.59% |

**First-degree burns have the lowest recall; Third-degree precision is 73.00%.**

# Classification model

## Interpretation and limitations

Internal test performance does not establish real-world reliability

### What the results show

- Macro-F1 reaches 85.59% on the internal test.
- First-degree recall is lower than other classes.
- No Burn is perfect on this split only.

### What remains uncertain

- Healthy skin was occasionally classified as First-degree outside the dataset.
- Image provenance and license records are incomplete. Duplicate screening does not guarantee a leakage-free split.

**Evaluate independent image sources, inspect First-degree errors, and strengthen traceability and duplicate checks across splits.**

# Segmentation model

## Dataset

Combine manually annotated masks, quality-checked pseudo-masks and healthy skin

| Mask source | Train manifest | Train used | Validation | Test |
|---|---|---|---|---|
| Manual polygon | 300 | 300 | 21 | 13 |
| QC pseudo-mask | 921 | 921 | 0 | 0 |
| None / background | 977 | 879 | 0 | 0 |
| Total | 2,198 | 2,100 | 21 | 13 |

**98 None images held out to monitor false-positive masks**

300 manual training records ≈ 100 original image groups with augmentation variants. Validation and test contain only manual masks; both sets are small.

# Segmentation model

## Mask preparation and model output

Standardize image quality and labels before fine-tuning

### Build training targets

- Manual polygons → binary masks
- Baseline U-Net++ → pseudo-masks
- Auto-QC: confidence, area, consistency
- Reject or isolate unreliable pseudo-masks

### Preprocessing & predict

- RGB images resized to 320 × 320
- ImageNet normalization
- Nearest-neighbor mask resizing
- SHA-256, pHash and group checks

- The mask identifies suspected burn regions; it does not assign burn depth per pixel.
- Empty or skipped annotations must not be treated as healthy skin labels.

**Output: one-channel prediction → binary mask → visual overlay**

## Training setup

# Segmentation model

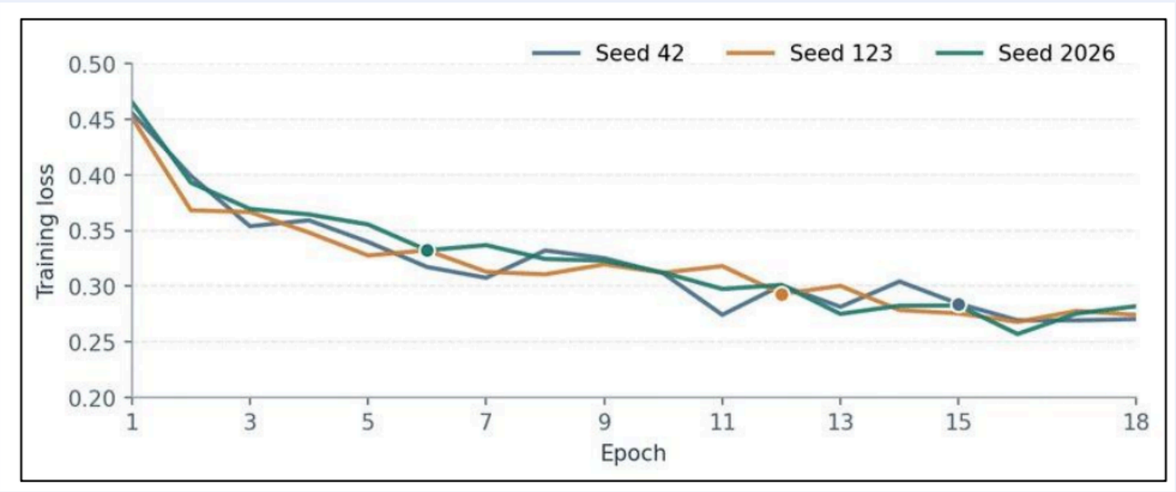
| Technical Parameter | Configured Value |
|---|---|
| Model Architecture | U-Net++ (Nested U-Net), Encoder ResNet-34 |
| Initial Weights | ImageNet Pretrained Weights |
| Input, Output Dimensions | 320 × 320 RGB, 1-channel binary logits |
| Batch Size & Steps | Batch size = 8; 80 steps/epoch (640 samples/epoch) |
| Maximum Epochs | 18 epochs |
| Loss Function | 0.5 × BCEWithLogits + 0.5 × SoftDice |
| Source Sampling in Train | 25% manual, 55% pseudo, 20% None |
| Checkpoint Criterion | SelectionScore = ValidDice - 0.15 × NegativeFPR |
| Best Checkpoint | best_random_curriculum_seed2026.pt (Epoch 6) |

640 sampled images per epoch, drawn with replacement

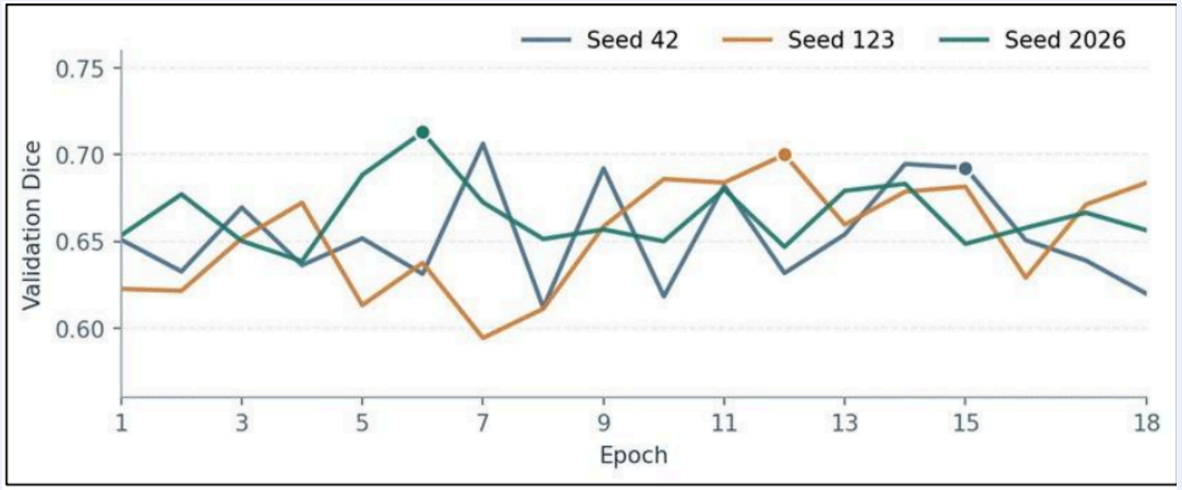
# Training Progress

Random Curriculum trained 18 epochs across seeds 42, 123 and 2026

## Segmentation model




*Training Loss curves by epoch for Random Curriculum across three seeds (42, 123, 2026)*




*Validation Dice variation by epoch across three seeds, with dots marking selected checkpoints*

## SEED 2026

**Training loss**

**0.4656 → 0.2819**

**Validation Dice fluctuates**

**65.34% → 65.64%**

**Lowest training loss**

**≠**

**best checkpoint**

# Segmentation model

## Checkpoint selection and false positives

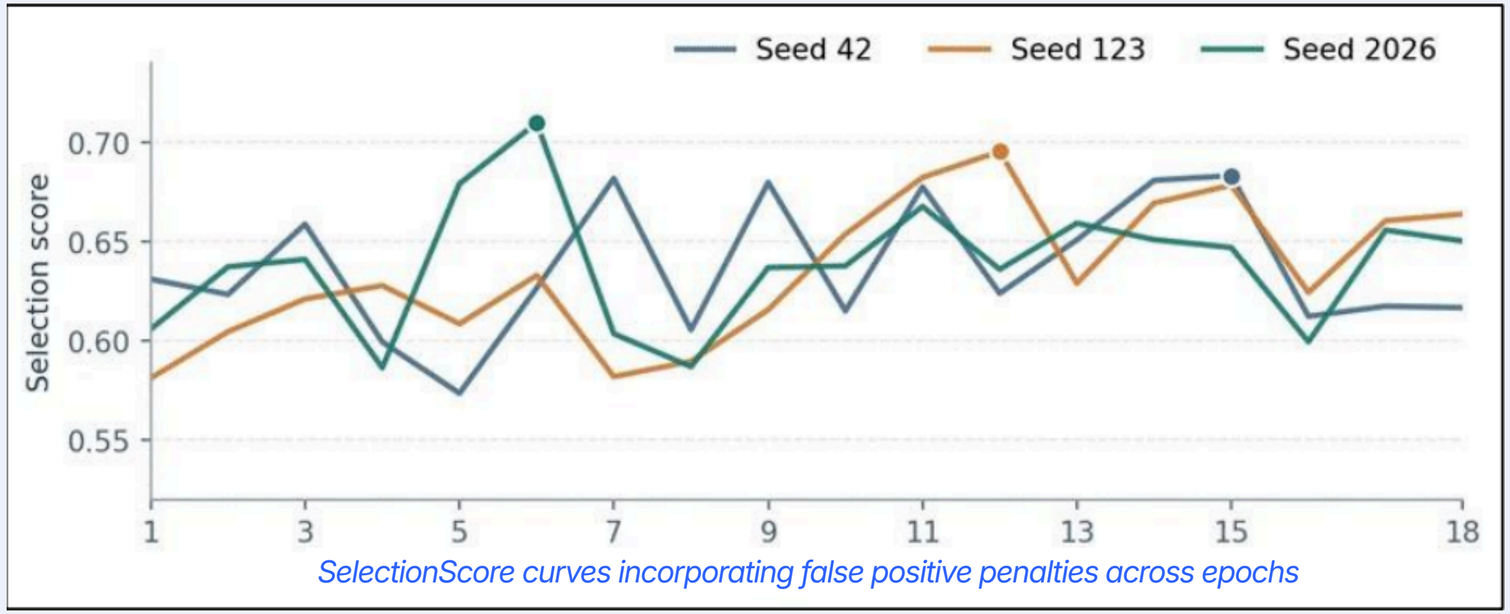
Balance overlap quality against false-positive masks on healthy skin

| Seed | Epoch | Val. Dice | NegativeFPR | Score |
|---|---|---|---|---|
| 42 | 15 | 69.24% | 6.12% | 0.6832 |
| 123 | 12 | 70.00% | 3.06% | 0.6954 |
| 2026 | 6 | 71.29% | 2.04% | 0.7099 |

**SelectionScore**
**=**
**Validation Dice − 0.15 × NegativeFPR**

**Selected: seed 2026, epoch 6**

- Validation IoU: 60.66%
- FPR measured on 98 held-out None images




*SelectionScore curves incorporating false positive penalties across epochs*

# Segmentation model

## Curriculum sampling strategy

Keep the source mix fixed while expanding the eligible positive image pool

## Four strategies evaluated

Uniform · Easy-to-Hard · Hard-to-Easy · Random Curriculum

- The pool reaches full size by approximately 75% of training.
- Pseudo-mask difficulty ranking showed no consistent benefit across seeds.

## 25% manual / 55% pseudo-mask / 20% None

Identical source sampling ratios across experiments

***Curriculum: 30% → 100% of eligible positive data***

## Curriculum vs. Manual-only baseline

Same preprocessing and evaluation metrics for both checkpoints

**Higher test Dice on 9 of 13 images**

| Evaluation | Metric | Manual-only v6 | Curriculum v5 |
|---|---|---|---|
| Validation | Mean Dice | 64.78% | 71.30% |
| Validation | Mean IoU | 53.07% | 60.67% |
| Test | Mean Dice | 65.92% | 72.15% |
| Test | Mean IoU | 54.64% | 60.14% |

# Segmentation model

## Interpretation and limitations

Internal test performance does not establish real-world reliability

### What the results show

- Binary suspected-burn mask
- Test Mean Dice: 72.15%
- Test Mean IoU: 60.14%
- Manual test set: 13 images

### What remains Limited

- Small manual validation and test sets
- Uneven annotation boundary precision
- No burn-degree prediction per pixel
- No quantitative external validation
- None labels are not clinical pixel masks

**Next steps: expand specialist-verified masks and external evaluation**

# Experimental Model Selection

| Criterion | Classification | Segmentation |
|---|---|---|
| Candidates | 5 CNN/Transformer architectures | 4 decoder/encoder architectures |
| Evaluation Set for Selection | 462 validation images, 4 classes | 21 manual validation masks + 98 holdout None images |
| Input, Batch Size | 224 × 224, 32 | 320 × 320, 8 |
| Training Budget | 15 epochs per seed | 18 epochs × 80 steps/epoch per seed |
| Seeds | 42, 123, 2026 | 42, 123, 2026 |
| Selection Metric | Validation Macro-F1 | Validation Dice - 0.15 × NegativeFPR |
| Initialization | Pretrained (timm) | Pretrained encoder, random curriculum |

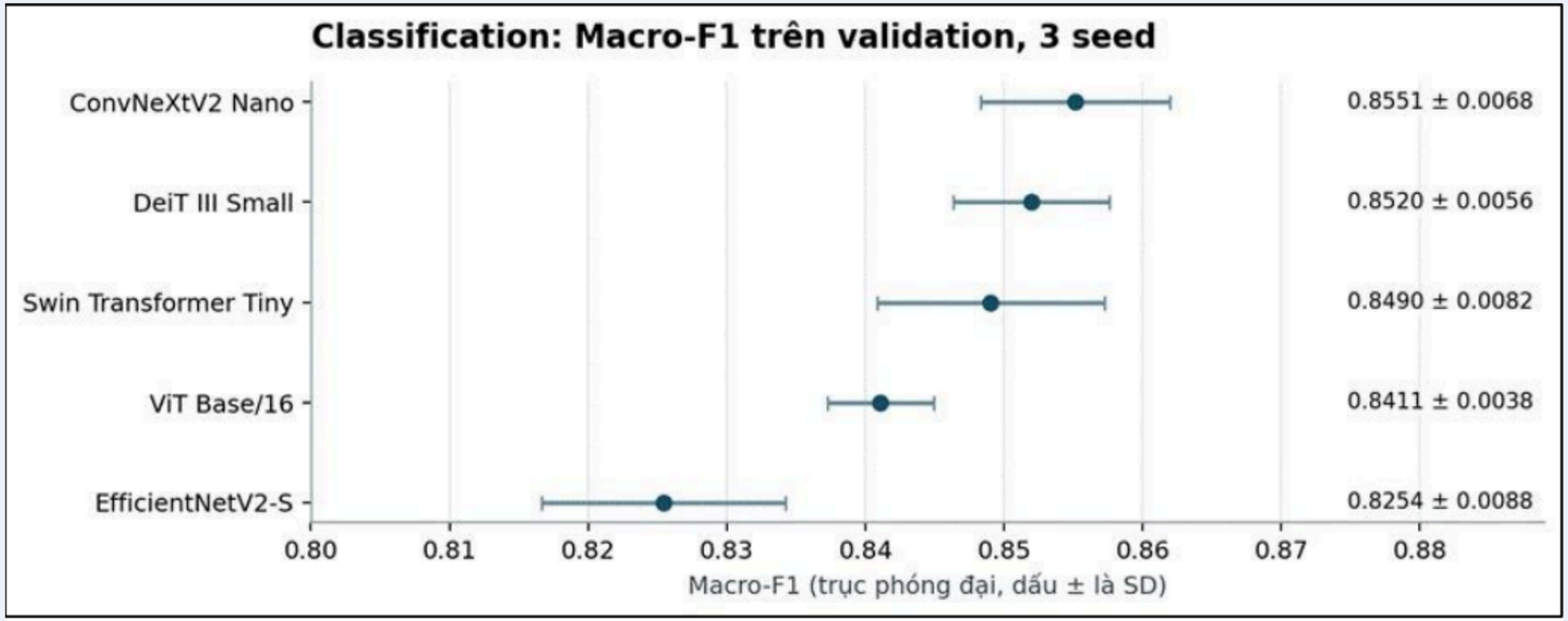
- **Objective:** Select architectures through controlled experiments and validation performance.
- **27 training runs:** 5 classification models + 4 segmentation models × 3 seeds (42, 123, 2026).
- **Hardware:** NVIDIA A100-SXM4-40GB GPU.
- Within each task, models shared the same data splits, training budget, and checkpoint criteria.
- Results were reported as mean ± standard deviation across seeds.

**Selection criteria:** Validation Macro-F1 for classification; Validation Dice – 0.15 × NegativeFPR for segmentation.

# Classification Architecture Selection

## Decision

Select ConvNeXtV2 Nano for its leading average performance and compact size.



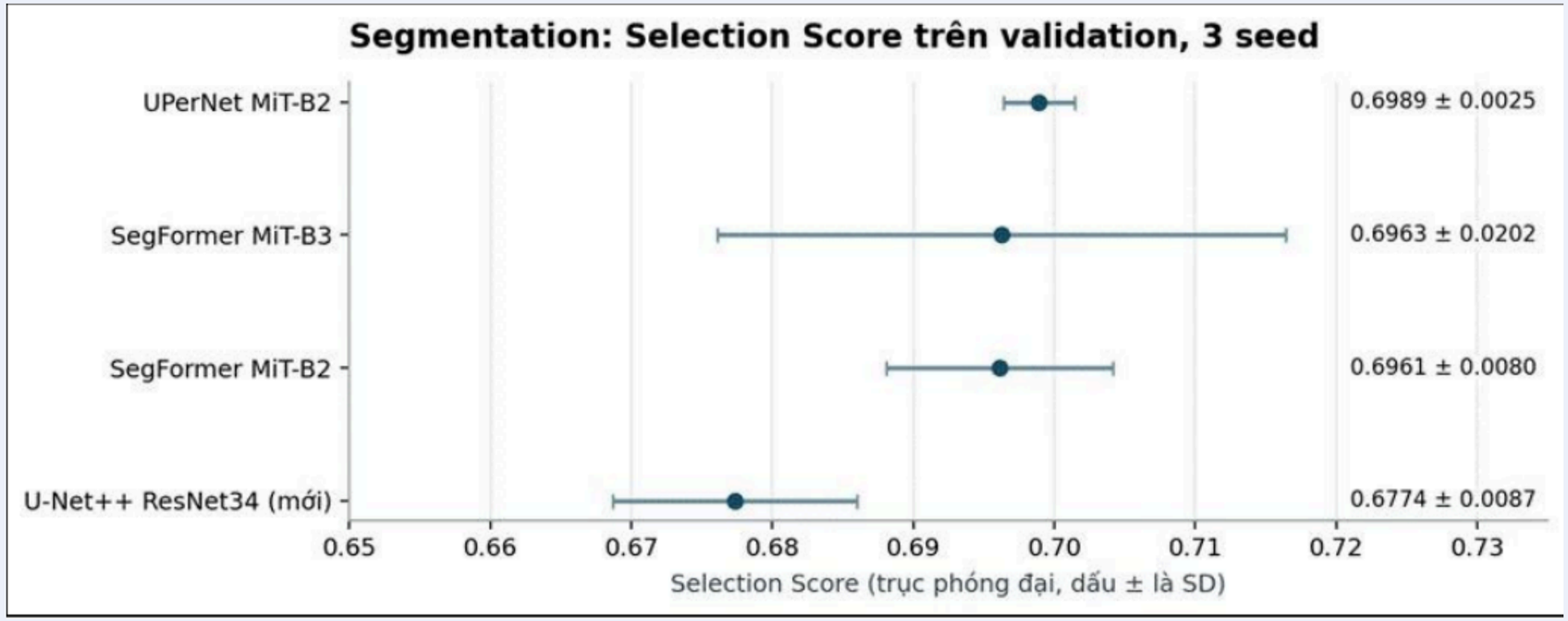

*Average Macro-F1 and standard deviation across three seeds; the horizontal axis is magnified to show small differences*

- Compared **ConvNeXtV2 Nano, DeiT III Small, Swin Tiny, ViT Base/16, and EfficientNetV2-S.**
- **ConvNeXtV2 Nano ranked first:** validation Macro-F1 85.51% ± 0.68 percentage points.
- DeiT III Small (85.20%) and Swin Tiny (84.90%) achieved similar average scores.
- ConvNeXtV2 also had the fewest parameters: 14.99M.

# Classification Architecture Selection

## Finding

UPerNet offered the strongest validation balance of overlap quality, false-positive control, and consistency across seeds.




*Average validation Selection Score and standard deviation across three seeds; horizontal axis is magnified to distinguish architectures*

- Compared **U-Net++ ResNet-34, SegFormer MiT-B2, SegFormer MiT-B3, and UPerNet MiT-B2** using Random Curriculum.
- **UPerNet MiT-B2 ranked first:** validation SelectionScore 0.6989 ± 0.0025.
- It achieved 0% NegativeFPR on 98 held-out None images at all three selected checkpoints.
- SegFormer MiT-B2 had slightly higher average Dice (70.02%), but false positives reduced its SelectionScore.

# Final model decision

- **Classification:** Selected **ConvNeXtV2 Nano** for the highest average validation Macro-F1 across five architectures and compact parameter count.
- **Segmentation:** Retained **Curriculum U-Net++ ResNet-34** (random_curriculum, seed 2026) for slightly higher test Mean Dice, zero empty masks across 13 test images, and a verified demo pipeline.
- **Alternative:** Preserved UPerNet MiT-B2 as a strong candidate due to minimal Mean Dice difference and superior performance on several other metrics.

ConvNeXtV2 predicts image-level labels; U-Net++ generates binary masks for visualization and contradiction checking.

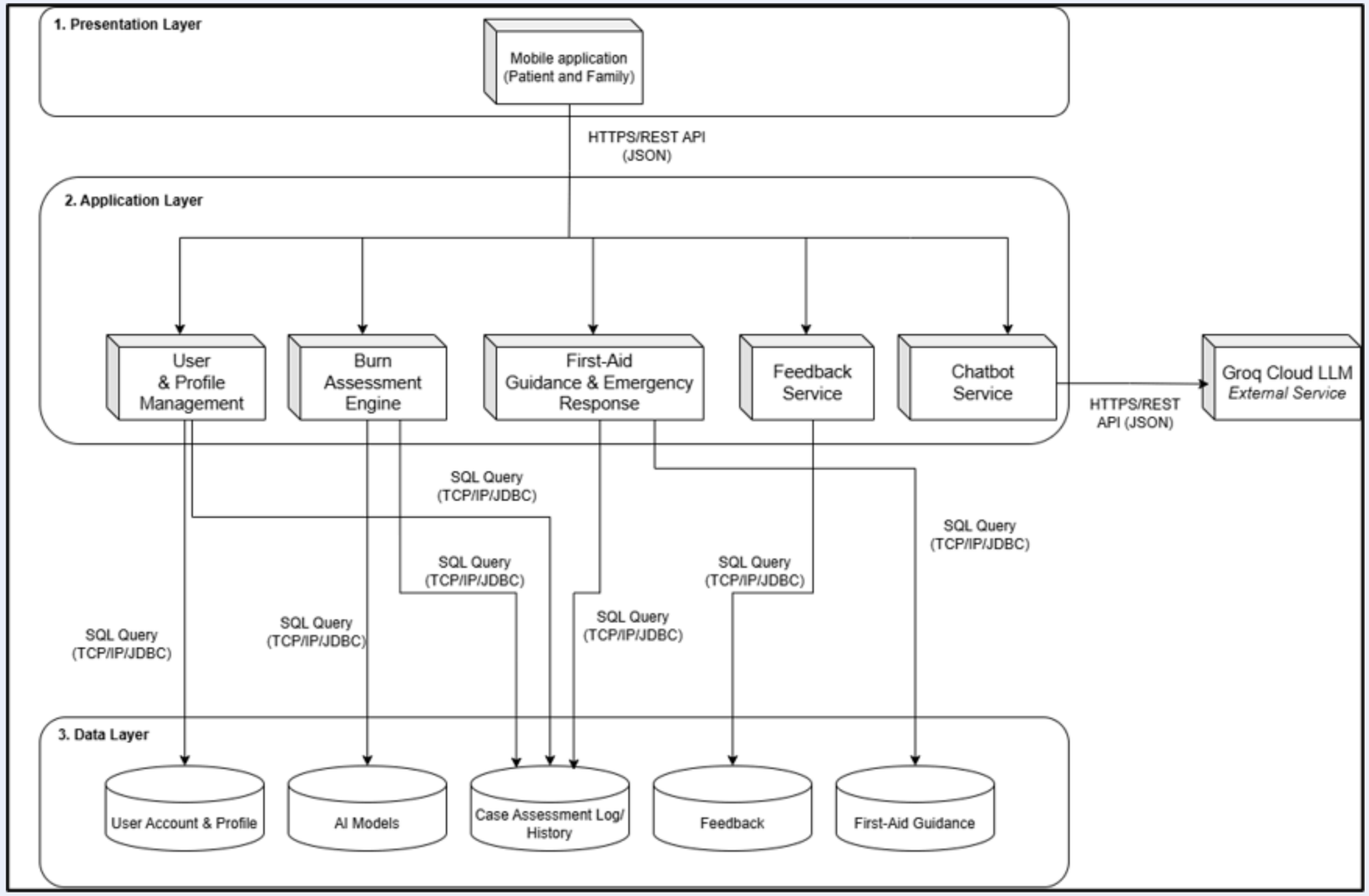
# Large language model & RAG

*Large Language Model (LLM) combined with Retrieval-Augmented Generation (RAG) serves as an intelligent medical assistant.*

- **LLM:** Open-source LLaMA 3.3 70B Versatile, deployed via Groq API, with millisecond response times.
- **Knowledge base:** 18 reputable textbooks and professional guidelines, including Military Medical Academy and Ministry of Health documents.
- **Data processing:** 2,274 text chunks stored in knowledge_base_burns_full.json.
- **Retrieval:** BM25 and vector embeddings retrieve 3–5 relevant chunks to construct LLM prompt context.

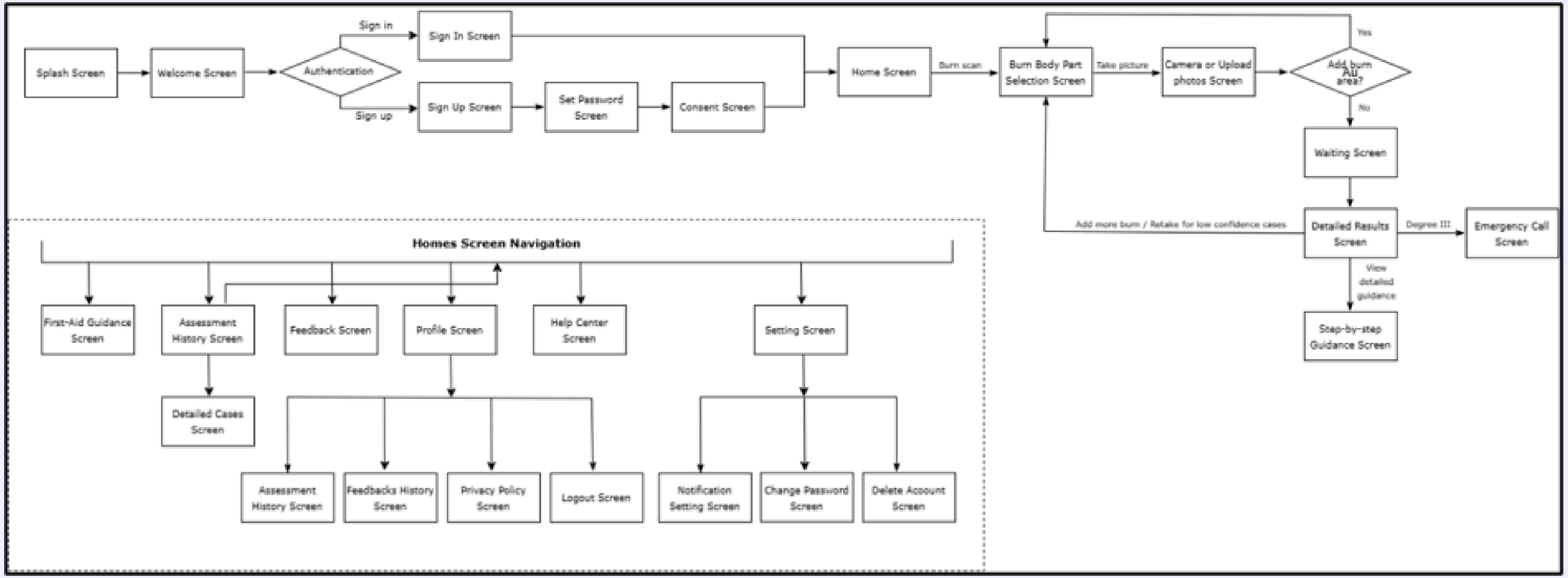
**Medical accuracy: Removed unverified professional titles, such as “board-certified,” from system prompts.**

# 3-tier architecture diagram




Burn Aid

# Navigation flowchart



# UI design




Burn Aid

# UI design

## Demo

# System Limitations



## Clinical scope

- Thermal burns caused by fire only;
- Not yet an age-adjusted for children under 18;



## Features

- One photo per assessment;
- Local SQLite; no doctor review screen or location sharing.



## AI and imaging

- Performance of models is affected by the quality of images taken in real situations
- Tiny scattered burns missed; chatbot needs internet.

# Short-Term Plan

1. Multi-injury assessment
2. Lund–Browder chart for under-18s
3. Doctor review of low-confidence cases
4. Chemical and electrical first aid

# Medium & Long-Term Plan

1. Local clinical dataset, Vietnamese skin
2. On-device chatbot for offline use
3. GIS: nearest facility, pre-hospital info

# Thank You


09:45
Choose Burn Body Part
Front View
Back View
TAP ON A NUMBERED REGION
SELECTED REGION
Forearm, Front, Right
Prof. Le The Trung Standard:
1.5% TBSA base for this view
ESTIMATED TBSA FOR REGION
0.75% TBSA
Burn affected extent in region:
50%
0%
20%
40%
60%
80%
100%
Reset Selection
Take Picture Now